

Lecture 15: Markov Chains and Multi Step Transitions

APRIL 1

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15.1 Review of Markov Chains

Example 15.1. *Consider a gambling game in which on any turn you win \$1 with probability $p = 0.4$ or lose \$1 with probability $1 - p = 0.6$. Suppose further that you adopt the rule that you quit playing if your fortune reaches $\$N$. Of course, if your fortune reaches $\$0$ the casino makes you stop.*

Definition 15.2. *Let X_n be the amount of money you have after n plays. Your fortune, X_n has the “Markov property.” In words, this means that given the current state, X_n , any other information about the past is irrelevant for predicting the next state X_{n+1} .*

Definition 15.3. *We say that X_n is a discrete time Markov chain with transition matrix $p(i, j)$ if*

$$\mathbb{P}(X_{n+1} = j | X_n = i) = p(i, j)$$

Example 15.4. *Describing the Random Walk in terms of a probability matrix.*

Example 15.5. *Describing the Gambler’s ruin in terms of a probability matrix.*

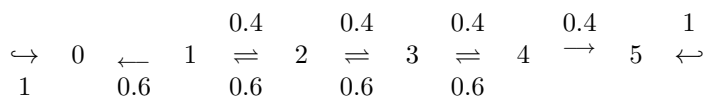
$$\begin{aligned} p(0, 0) &= 1 \\ p(N, N) &= 1 \\ p(i, i+1) &= 0.4, \quad 0 < i < N \\ p(i, i-1) &= 0.6, \quad 0 < i < N \end{aligned}$$

if we make a matrix out of this we have

	0	1	2	3	4	5
0	1.0	0	0	0	0	0
1	0.6	0	0.4	0	0	0
2	0	0.6	0	0.4	0	0
3	0	0	0.6	0	0.4	0
4	0	0	0	0.6	0	0.4
5	0	0	0	0	0	1.0

We can also view this as a collection of states and ways to move between them.

Example 15.6. *This chain originated in physics as a model for two cubical volumes of air connected by a small hole. In the mathematical version, we have two “urns,” i.e., two of the exalted trash cans of probability*



theory, in which there are a total of N balls. We pick one of the N balls at random and move it to the other urn. Let X_n be the number of balls in the “left” urn after the n^{th} draw. It should be clear that X_n has the Markov property; i.e., if we want to guess the state at time $(n+1)$, then the current number of balls in the left urn X_n , is the only relevant information from the observed sequence of states $X_n, X_{n-1}, \dots, X_1, X_0$. To check this we note that

$$P(X_{n+1} = i+1 | X_n = i) = \frac{N-i}{N},$$

since to increase the number we have to pick one of the $N-i$ balls in the other urn. The number can also decrease by 1 with probability i/N . In symbols, we have computed that the transition probability is given by

$$p(i, i+1) = \frac{N-i}{N} \quad p(i, i-1) = \frac{i}{N}.$$

When $N = 4$, we have

	0	1	2	3	4
0	0	1.0	0	0	0
1	1/4	0	3/4	0	0
2	0	1/2	0	1/2	0
3	0	0	3/4	0	1/4
4	0	0	0	1.0	0

One more commonly describes a Markov chain by writing down a transition probability $p(i, j)$ with

1. $p(i, j) \geq 0$, since they are probabilities.
2. $\sum_j p(i, j) = 1$, since when $X_n = i$, X_{n+1} will be in some state j .

The equation is read “sum $p(i, j)$ over all possible values of j .” In words the last two conditions say: the entries of the matrix are nonnegative and each ROW of the matrix sums to 1.

1. Let X_n be the weather on day n in Ithaca, NY, which we assume is either: 1 = rainy, or 2 = sunny. Even though the weather is not exactly a Markov chain, we can propose a Markov chain model for the weather by writing down a transition probability. The table says, for example, the probability a rainy

	1	2
1	0.6	0.4
2	0.2	0.8

day (state 1) is followed by a sunny day (state 2) is $p(1, 2) = 0.4$. A typical question of interest is: What is the long-run fraction of days that are sunny?

Example 15.7. Let X_n be a family’s social class in the n^{th} generation, which we assume is either 1 = lower, 2 = middle, or 3 = upper. In our simple version of sociology, changes of status are a Markov chain with the following transition probability

Do the fractions of people in the three classes approach a limit?

	1	2	3
1	0.7	0.2	0.1
2	0.3	0.5	0.2
3	0.2	0.4	0.4

Example 15.8. Suppose there are three types of laundry detergent, 1, 2, and 3, and let X_n be the brand chosen on the n^{th} purchase. Customers who try these brands are satisfied and choose the same thing again with probabilities 0.8, 0.6, and 0.4 respectively. When they change they pick one of the other two brands at random. The transition probability is Do the market shares of the three product stabilize?

	1	2	3
1	0.8	0.1	0.1
2	0.2	0.6	0.2
3	0.3	0.3	0.4

Example 15.9. In a Markov chain the distribution of X_{n+1} only depends on X_n . This can easily be generalized to case in which the distribution of X_{n+1} only depends on (X_n, X_{n-1}) . For a concrete example consider a basketball player who makes a shot with the following probabilities:

- 1/2 if he has missed the last two times
- 2/3 if he has hit one of his last two shots
- 3/4 if he has hit both of his last two shots

To formulate a Markov chain to model his shooting, we let the states of the process be the outcomes of his last two shots: $\{HH, HM, MH, MM\}$ where M is short for miss and H for hit. The transition probability is

	HH	HM	MH	MM
HH	3/4	1/4	0	0
HM	0	0	2/3	1/3
MH	2/3	1/3	0	0
MM	0	0	1/2	1/2

To explain suppose the state is HM , i.e., $X_{n-1} = H$ and $X_n = M$. In this case the next outcome will be H with probability 2/3. When this occurs the next state will be $(X_n, X_{n+1}) = (M, H)$ with probability 2/3. If he misses an event of probability 1/3, $(X_n, X_{n+1}) = (M, M)$.

15.2 Multistep Transition Probabilities

The transition probability $p(i, j) = \mathbb{P}(X_{n+1} = j | X_n = i)$ gives the probability of going from i to j in one step. Our goal in this section is to compute the probability of going from i to j in $m > 1$ steps:

$$p^m(i, j) = \mathbb{P}(X_{n+m} = j | X_n = i)$$

To warm up, we recall the transition probability of the social mobility chain:

	Lower	Middle	Upper
Lower	0.7	0.2	0.1
Middle	0.3	0.5	0.2
Upper	0.2	0.4	0.4

Question 1. Your parents were middle class (state 2). What is the probability that you are in the upper class (state 3) but your children are lower class (state 1)?

Solution 1. Intuitively, the Markov property implies that starting from state 2 the probability of jumping to 3 and then to 1 is given by

$$p(2,3)p(3,1)$$

To get this conclusion from the definitions, we note that using the definition of conditional probability,

$$\begin{aligned}
 \mathbb{P}(X_2 = 1, X_1 = 3 | X_0 = 2) &= \frac{\mathbb{P}(X_2 = 1, X_1 = 3, X_0 = 2)}{\mathbb{P}(X_0 = 2)} \\
 &= \frac{\mathbb{P}(X_2 = 1, X_1 = 3, X_0 = 2)}{\mathbb{P}(X_1 = 3, X_0 = 2)} \frac{\mathbb{P}(X_1 = 3, X_0 = 2)}{\mathbb{P}(X_0 = 2)} \\
 &= \mathbb{P}(X_2 = 1 | X_1 = 3, X_0 = 2) \mathbb{P}(X_1 = 3 | X_0 = 2) \\
 &= \mathbb{P}(X_2 = 1 | X_1 = 3) \mathbb{P}(X_1 = 3 | X_0 = 2) = p(2,3)p(3,1)
 \end{aligned}$$

Question 2. What is the probability your children are lower class (1) given your parents were middle class (2)?

Solution 2. To do this we simply have to consider the three possible states for your class and use the solution of the previous problem

$$\begin{aligned}
 \mathbb{P}(X_2 = 1 | X_0 = 2) &= \sum_{k=1}^3 \mathbb{P}(X_2 = 1, X_1 = k | X_0 = 2) = \sum_{k=1}^3 p(2,k)p(k,1) \\
 &= (.3)(.7) + (.5)(.3) + (.2)(.2) = .21 + .15 + .04 = .21
 \end{aligned}$$

There is nothing special here about the states 2 and 1 here. By the same reasoning,

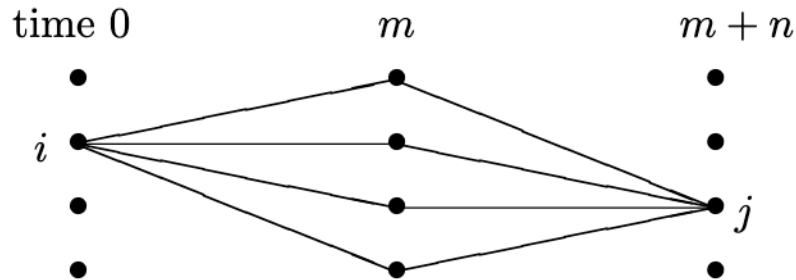
$$\mathbb{P}(X_2 = j | X_0 = i) = \sum_{k=1}^3 p(i,k)p(k,j)$$

The right-hand side of the last equation gives the $(i,j)^{th}$ entry of the matrix p is multiplied by itself. To explain this, we note that to compute $p^2(2,1)$ we multiplied the entries of the second row by those in the first column:

$$\begin{pmatrix} . & . & . \\ 0.3 & 0.5 & 0.2 \\ . & . & . \end{pmatrix} \begin{pmatrix} 0.7 & . & . \\ 0.3 & . & . \\ 0.2 & . & . \end{pmatrix} = \begin{pmatrix} . & . & . \\ 0.40 & . & . \\ . & . & . \end{pmatrix}$$

If we wanted $p^2(1,3)$ we would multiply the first row by the third column

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ . & . & . \\ . & . & . \end{pmatrix} \begin{pmatrix} . & . & 0.1 \\ . & . & 0.2 \\ . & . & 0.4 \end{pmatrix} = \begin{pmatrix} . & . & 0.15 \\ . & . & . \\ . & . & . \end{pmatrix}$$



When all of the computations are done we have

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix} \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix} = \begin{pmatrix} 0.57 & 0.28 & 0.15 \\ 0.40 & 0.39 & 0.21 \\ 0.34 & 0.40 & 0.26 \end{pmatrix}$$

Theorem 15.10. The m step transition probability $\mathbb{P}(X_{n+m} = j | X_n = i)$ is the m^{th} power of the transition matrix p .

The key ingredient is proving the **Chapman-Kolmogorov equation**

$$p^{m+n}(i, j) = \sum_k p^m(i, k) p^n(k, j).$$

Why is (1.2) true? To go from i to j in $m + n$ steps, we have to go from i to some state k in m steps and then from k to j in n steps. The Markov property implies that the two parts of our journey are independent

Example 15.11 (Gambler's Ruin). Suppose for simplicity that $N = 4$ in Example 1.1, so that the transition probability is

	0	1	2	3	4
0	1.0	0	0	0	0
1	0.6	0	0.4	0	0
2	0	0.6	0	0.4	0
3	0	0	0.6	0	0.4
4	0	0	0	0	1.0

Compute p^2 .